

Flooring Subfloor Build Process

Scope

- Define the locked floor stack-up, hardpoint strategy, procurement state, and install workflow for the truck bed floor.
- Baseline date for this revision: May 14, 2026.
- Moisture-control premise updated May 3, 2026: the Hiatus is effectively a sealed capper with barn doors; the inside of the truck bed remains the build substrate, so water must not be allowed under the subfloor.
- Field progress updated May 4, 2026: bed rail caps were re-bonded/sealed with polyurethane; small drill holes and corner gaps were patched/sealed; trowel is purchased.
- Field progress updated May 14, 2026: the subfloor is complete enough to serve as the current working datum, while Lonseal remains intentionally unglued until rough-in, hardpoints, and service-access decisions are validated in the installed camper. Owner update 2026-06-01: sheet vinyl/Lonseal installation is expected late, after interior buildout/service-access gates, not as a near-term task.

Final Subfloor Structure (Bottom to Top)

1. Existing spray-in bedliner.
2. 5/8 in SilveRboard Graphite EPS between ribs only. Current state: installed/trimmed enough for the current subfloor datum; inspect for proud spots, rocking, moisture paths, or hardpoint conflicts before finish-floor closure.
3. 3/4 in birch plywood directly over EPS + rib structure. Current state: complete in three slices with bottom/edges sealed and top bond face untreated for Lonseal adhesive; keep removable/serviceable until rough-in and hardpoint gates pass.
4. Lonseal Lonwood Madera marine sheet vinyl, glue-down finish layer. Current state: sheet vinyl and adhesive are in hand; glue-down is a late finish step after interior buildout/rough-in gates because it reduces subfloor serviceability.

Explicitly De-Scoped Layer

- Full-bed EPDM/RPDB thermal-break sheet above EPS and below plywood is no longer in the active floor stack-up.
- EPDM order was canceled before shipment and refunded; no EPDM material is on hand for this build.

Hardpoint Rule (Structural Anchors)

- At structural hardpoints, remove EPS locally so plywood bears directly on steel.
- Use mechanical fasteners into rivnuts at these pockets.
- Do not place compressible insulation in bolt clamp stacks.

Procurement Snapshot (Updated)

Purchased / in hand (active flooring path)

- SilveRboard Graphite EPS kit (\$71.38) - Home Depot.
- Loctite PL 300 foamboard adhesive (\$5.48) - Home Depot.
- TotalBoat Halcyon Water-Based Marine Varnish, clear gloss, pint (\$22.99) - Amazon.
- 3/4 in birch plywood, 4x8 , qty 2 (\$88 each, \$176 total).
- Lonseal Lonwood Madera Topseal marine sheet vinyl.
- Lonseal adhesive.
- Sheet-vinyl adhesive trowel: Roberts 10-181-16 exact-notch trowel (\$4.40 ; BOM row 174).

Purchased / partially used (bed sealing workstream)

- Loctite polyurethane window/door/siding sealant, 10 oz , qty 2 at \$9 each (\$18 ; BOM row 177), used under bed rail caps and at selected corner gaps.
- Gorilla Waterproof Patch & Seal Tape, 8 ft roll (\$14 ; BOM row 178), roughly half used on small drill holes and other minor bed openings.

Canceled / refunded (not in hand)

- Rubber-Cal EPDM closed-cell sponge rubber, qty 2 (\$47.22) - Home Depot order canceled before shipment; refunded.

Related purchase (not floor stack, but same workstream)

- 1/2 in plywood, 4x8 , qty 1 (\$68) for electrical backer/closet module.

Not Yet Purchased / confirm before glue-down

- Compact segmented roller or equivalent if not already in hand (row 175).

- Fresh utility blades, masking tape, gloves, rags/cleaner, and disposable adhesive cleanup supplies as needed.

Totals

- Active flooring package purchased to date: \$280.25 (adds purchased trowel row 174).
- Active flooring package remaining expected spend: \$424.68 .
- Active flooring package estimated total: \$704.93 .
- Bed sealing consumables added outside the prior flooring package total: \$32 (rows 177-178).

Historical provenance

- [INSTALL MINUS 12 READINESS PLAN](#) preserves the May 7 install-window floor/seal planning.
- [STARTER PLAN electrical and flooring pre camper](#) preserves the pre-camper floor/electrical starter plan.

Installation Sequence

Phase 0 - Bed rail dust/weather closeout before finish floor

Status: historical install-window workstream; inspect after normal travel/shakedown before any finish-floor closure.

1. Empty the bed and remove the existing removable flooring before the Washington trip; do not transport flooring through the wet/adhesive rail-seal workstream. **Completed.**
2. Remove or lift bed rail caps as needed and photograph all old camper-install holes. **Caps removed; most clips broke.**
3. Consider tailgate removal only if it improves access or drying; store it safely and protect wiring/camera connections.
4. Confirm with Hiatus that the old roughly 2 in x 3 in top-rail holes are not needed for the new camper interface if still relevant to the final attachment method.
5. Clean/degrease rail surfaces and rust-protect bare cut edges.
6. Close holes as dust/weather intrusion points using low-profile, serviceable cover patches where possible: thin aluminum/ABS/HDPE or similar non-absorbent patch/backer bedded in butyl tape, automotive seam sealer, MS polymer, or polyurethane sealant.
7. Reinstall rail caps with targeted sealing/gasketing while avoiding camper-fit interference and avoiding hidden pockets that can hold water against metal. **Completed with large Loctite polyurethane bead under caps.**
8. Let sealant skin/cure per product conditions before reinstalling flooring or handling surfaces. **Historical gate; perform post-travel inspection before closeout.**

Phase 0A - Bed-floor drain/corner sealing before subfloor closeout

Status: partially complete as of May 4; cure/post-drive inspection still required.

1. Treat factory drain holes, unused bed-floor penetrations, and small corner gaps under/near the subfloor as water-ingress defects, not intentional drains for this build.
2. Seal them before final subfloor/Lonseal closeout using reversible or serviceable automotive/marine methods where practical: OEM-style rubber plugs, butyl-bedded aluminum/ABS/HDPE patches, automotive seam sealer, MS polymer, polyurethane sealant, or patch/seal tape for small drill holes where adhesion is clean and protected.
3. Do not rely on household silicone as the primary sealant.
4. Do not seal unrelated exterior body drains or create trapped metal cavities; the target is zero upward water path into the EPS/plywood cavity, while still allowing inspection/service before the finish floor is glued.
5. Field status: small drill holes and several other minor bed openings were sealed with Gorilla Waterproof Patch & Seal Tape; selected corner air gaps received small polyurethane beads.
6. After sealant cure, camper install, and normal travel/shakedown, inspect the bed floor/corners and inside of the bed before reinstalling EPS/plywood. No dedicated hose/leak test is required for this under-subfloor path; the concern is ordinary underside spray/dust, not submersion.

Phase 1 - Prep / current-state correction

1. Clean bed thoroughly.
2. Trim EPS so it is not proud, buckled, or interfering with plywood seating.
3. Re-check structural hardpoints: remove EPS locally anywhere plywood must bear directly on steel.
4. Confirm the three plywood slices sit flat and can still be removed intentionally.

Phase 2 - Between-Rib Insulation (mostly complete)

1. EPS remains between ribs only.
2. Maintain light tack-only adhesion where used; avoid full-surface bonding.
3. Confirm EPS does not interfere with hardpoint clearance zones.

Phase 3 - Subfloor (3/4 in birch, three-slice state)

1. Verify panel seam strategy and removal path before finish flooring.
2. Confirm sealed bottom/edges/holes and untreated clean top bond face.
3. If new anchor holes are drilled, seal the inside of all plywood holes before final install.
4. Reinstall plywood and bolt structural hardpoints only after hardpoint pockets are clear.

Phase 4 - Lonseal glue-down gate

Do not proceed until:

- bed rail, bed-floor drain-hole, and corner-gap sealing is complete and visually inspected after cure/normal use, not merely deferred;
- EPS trim and plywood seating pass;
- no floor-through penetrations, drains, heater routes, anchor holes, or hidden hardpoints need access under the finish sheet;
- top plywood face is clean, dry, untreated, and adhesive-compatible;
- exact trowel is purchased and roller/rolling method is confirmed;
- dry cure window is available.

Phase 5 - Finish Flooring

1. Dry-fit Lonseal sheet and align grain/pattern.
2. Apply adhesive per Lonseal adhesive manufacturer specification using the exact required trowel.
3. Lay vinyl and roll thoroughly.
4. Trim edges and allow cure before loading furniture.

Final Build Height

- EPS: 0.625 in
- Plywood: 0.750 in
- Vinyl: 0.080 in
- Total: approximately 1.46 in

Confirm final door and tank clearances against this stack-up before permanent installation.

Engineering Notes

- Thermal performance: moderate floor thermal decoupling from EPS; less top-side thermal break than prior EPDM-inclusive concept.
- Structural behavior: stiffer walking surface and higher point-load robustness with 3/4 in birch.
- Moisture durability: sealed bottom face, edges, and drilled holes remain required because wood is directly above rib cavities; the top face stays untreated to remain compatible with the Lonseal vehicle-install substrate requirement for untreated plywood.
- Bed-floor water policy: the subfloor cavity is not a drained cavity in this build. Because the camper has sealed barn doors and the bed floor remains the build substrate, factory drain holes and small corner gaps that can admit water under the subfloor should be sealed and visually inspected before finish-floor glue-down; no dedicated hose/leak test is required for ordinary underside-spray risk.
- Acoustic behavior: no dedicated EPDM damping layer in active stack; expect more direct structure-borne vibration vs prior concept.
- Serviceability: finish vinyl remains the only permanent bond layer, but one-piece Lonseal glued over three plywood slices will make subfloor removal materially harder. Treat glue-down as an explicit gate, not casual finish work.